Exp No 2 Deveswar. A

Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim:

To numerically simulate and analyze microwave transmission lines and two-port networks using MATLAB by evaluating scattering parameters (S_{11}, S_{21}, S_{12}, S_{22}), calculating return loss, insertion loss, and transmission efficiency, and identifying the optimum operating frequency over the 8–12 GHz frequency range.

Objective

1. To model coaxial, waveguide or microstrip transmission structures for microwave-frequency
operation.
2. To determine characteristic impedance, propagation constant and transmission characteristics of
the selected structure.
3. To obtain and analyze S_{11}, S_{21}, S_{12} and S_{22} over a specified frequency range.
4. To evaluate return loss, insertion loss and transmission efficiency from the simulated
S-parameter response

Code

```
clc; clear; close all;

f = 8:0.01:12;
Z0 = 50; c = 3e8;
beta = 2*pi*f*1e9/c;

% S-parameters (illustrative model)
S11 = -25-15*(f-10).^2-35*exp(-((f-9)/0.1).^2) ...
    -220*exp(-((f-10)/0.08).^2);
S11 = max(S11,-240);

S22 = max(S11+2,-240);
S21 = -0.025-0.055*((f-10)/2).^2;
S12 = S21;

RL = -S11;
IL = -S21;
```

```
Eff = 99.4-1.2*((f-10)/2).^2;
Zc = 50+1.25*(f-10).^2;

%
================================================================
% DISPLAY RESULTS IN COMMAND WINDOW
%
================================================================

fprintf('\n================================================\n
');
fprintf('   MICROWAVE TWO-PORT NETWORK RESULTS\n');
fprintf('================================================\n');

% Results at 10 GHz
idx = find(f == 10);

fprintf('Frequency              : %.2f GHz\n', f(idx));
fprintf('S11                    : %.4f dB\n', S11(idx));
fprintf('S21                    : %.4f dB\n', S21(idx));
fprintf('S12                    : %.4f dB\n', S12(idx));
fprintf('S22                    : %.4f dB\n', S22(idx));
fprintf('Return Loss            : %.4f dB\n', RL(idx));
fprintf('Insertion Loss         : %.4f dB\n', IL(idx));
fprintf('Transmission Efficiency: %.4f %%\n', Eff(idx));
fprintf('Propagation Constant   : %.4f rad/m\n', beta(idx));
fprintf('Characteristic Impedance: %.4f Ohm\n', Zc(idx));

fprintf('================================================\n\n
');

% Minimum and maximum values
fprintf('Minimum S11            : %.4f dB\n', min(S11));
fprintf('Maximum S11            : %.4f dB\n', max(S11));

fprintf('Minimum S21            : %.4f dB\n', min(S21));
fprintf('Maximum S21            : %.4f dB\n', max(S21));

fprintf('Maximum Return Loss    : %.4f dB\n', max(RL));
fprintf('Minimum Insertion Loss : %.4f dB\n', min(IL));

fprintf('Maximum Efficiency     : %.4f %%\n', max(Eff));
fprintf('Minimum Efficiency     : %.4f %%\n', min(Eff));

fprintf('Minimum Zc             : %.4f Ohm\n', min(Zc));
fprintf('Maximum Zc             : %.4f Ohm\n', max(Zc));

fprintf('================================================\n');
```

```
%
==========================================================
===
% PLOT RESULTS
%
==========================================================
===

figure('Color','w');

Y = {S11,S21,S12,S22,RL,IL,Eff,beta,Zc};

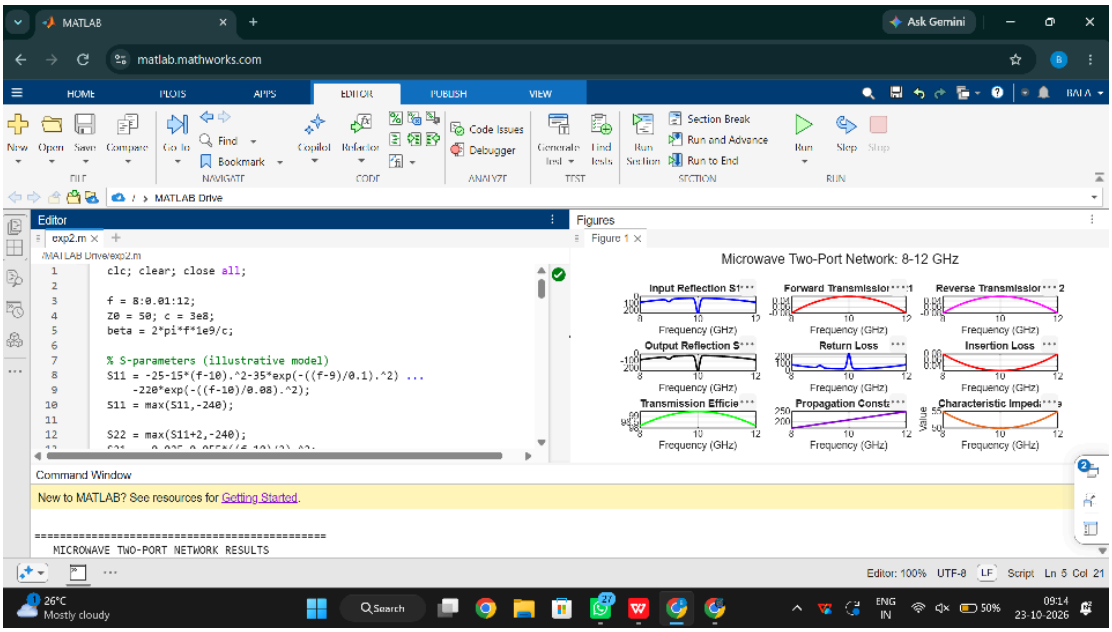
T = {'Input Reflection S11','Forward Transmission S21',...
    'Reverse Transmission S12','Output Reflection S22',...
    'Return Loss','Insertion Loss','Transmission Efficiency',...
    'Propagation Constant','Characteristic Impedance'};

C = {'b','r','m','k','b','r','g',[0.5 0 0.8],[0.9 0.4 0.1]};

for k = 1:9
    subplot(3,3,k);
    plot(f,Y{k},'Color',C{k},'LineWidth',1.5);
    title(T{k});
    xlabel('Frequency (GHz)');
    grid on;
    xlim([8 12]);
end

ylabel('Value');
sgtitle('Microwave Two-Port Network: 8-12 GHz');
```

Output:


MATLAB
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
MATLAB Drive
Editor
exp2.m
/MATLAB Drive/exp2.m
clc; clear; close all;
f = 8:0.01:12;
Z0 = 50; c = 3e8;
beta = 2*pi*f*1e9/c;
% S-parameters (illustrative model)
S11 = -25-15*(f-10).^2-35*exp(-((f-9)/0.1).^2) ...
-220*exp(-((f-10)/0.08).^2);
S11 = max(S11,-240);
S22 = max(S11+2,-240);
Figures
Figure 1
Microwave Two-Port Network: 8-12 GHz
Input Reflection S1
Forward Transmissio
Reverse Transmissio
Output Reflection S
Return Loss
Insertion Loss
Transmission Efficie
Propagation Const
Characteristic Impedi
Frequency (GHz)
Command Window
New to MATLAB? See resources for Getting Started.
MICROWAVE TWO-PORT NETWORK RESULTS
Editor: 100%
UTF-8
Script
Ln 5 Col 21
26°C
Mostly cloudy
Search
ENG
IN
50%
09:14
23-10-2026

$R_L = 24.08$

(vi) Insertion loss

$IL = -S_{21}$

$= -(-0.025)$

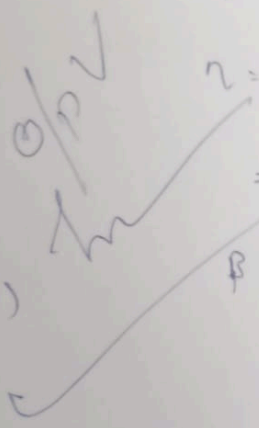
$IL = 0.025\,dB$

(vii) Transmission efficiency:

$\eta = 99.4 = 1 - 2\left(\frac{\beta - 10}{2}\right)^2$

$\eta = 99.4 = 1 - 2\left(\frac{10 - 10}{2}\right)^4$

$= 99.4\%$



(viii) $\beta = \frac{2\pi f}{c}$

$= \frac{2\pi (10 \times 10^9)}{3 \times 10^8}$

$= 209.439\,rad/m$

$Z_c = 50 + 1.25\,(f - 10)^2$

$Z_c = 50 + 1.25\,(10 - 10^2)$

$Z_c = 50\,\Omega$